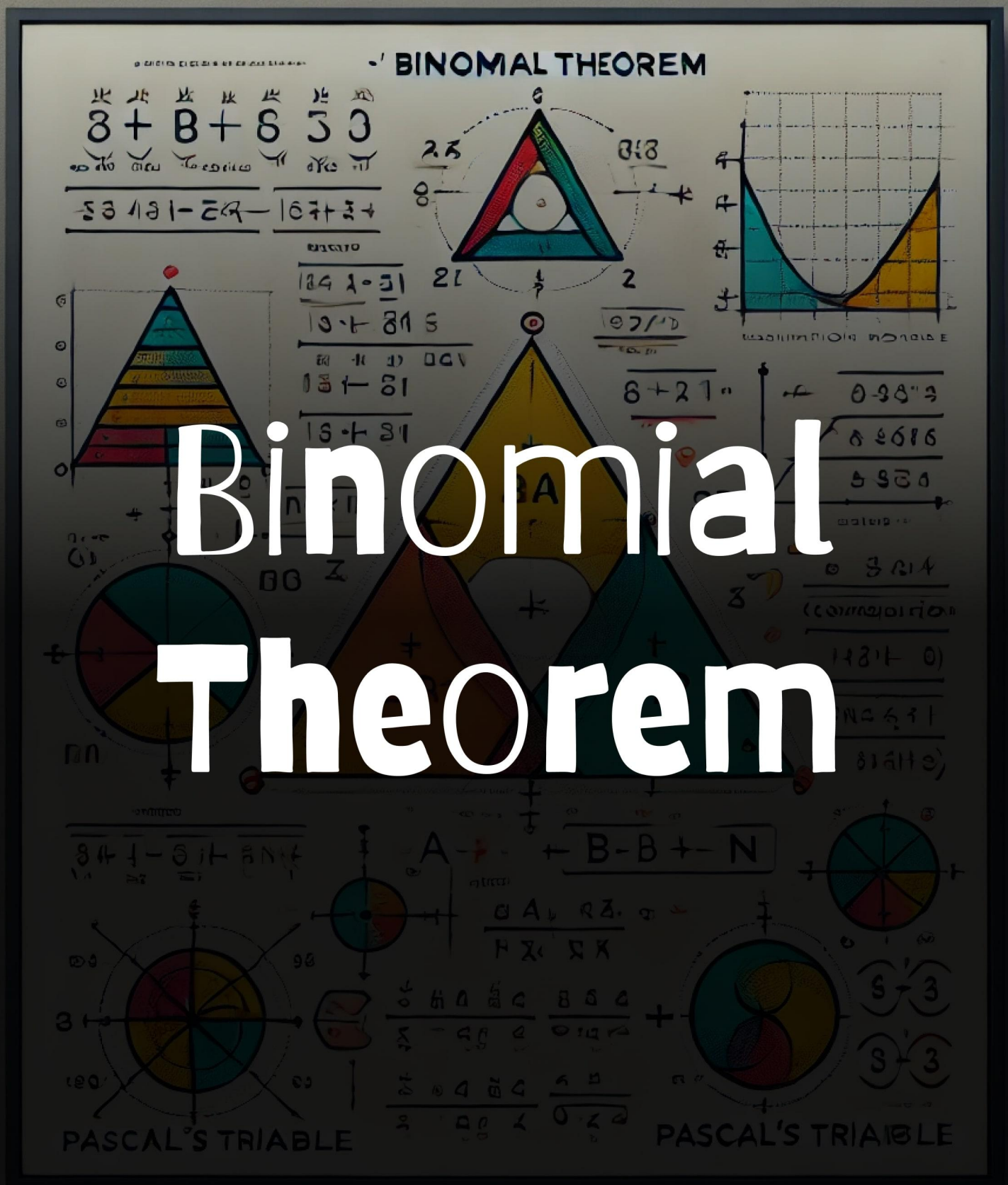



# Binomial Theorem

# BiNoMiaL TheoReM

LEARNING :)

FUN!

$x+y$ , $x+\frac{1}{x}$ , $(x+y)^2$ , $(x+y)^3$ , $\left(x-\frac{1}{2x^2}\right)^n$

## Binomials

### BINOMIAL EXPANSION

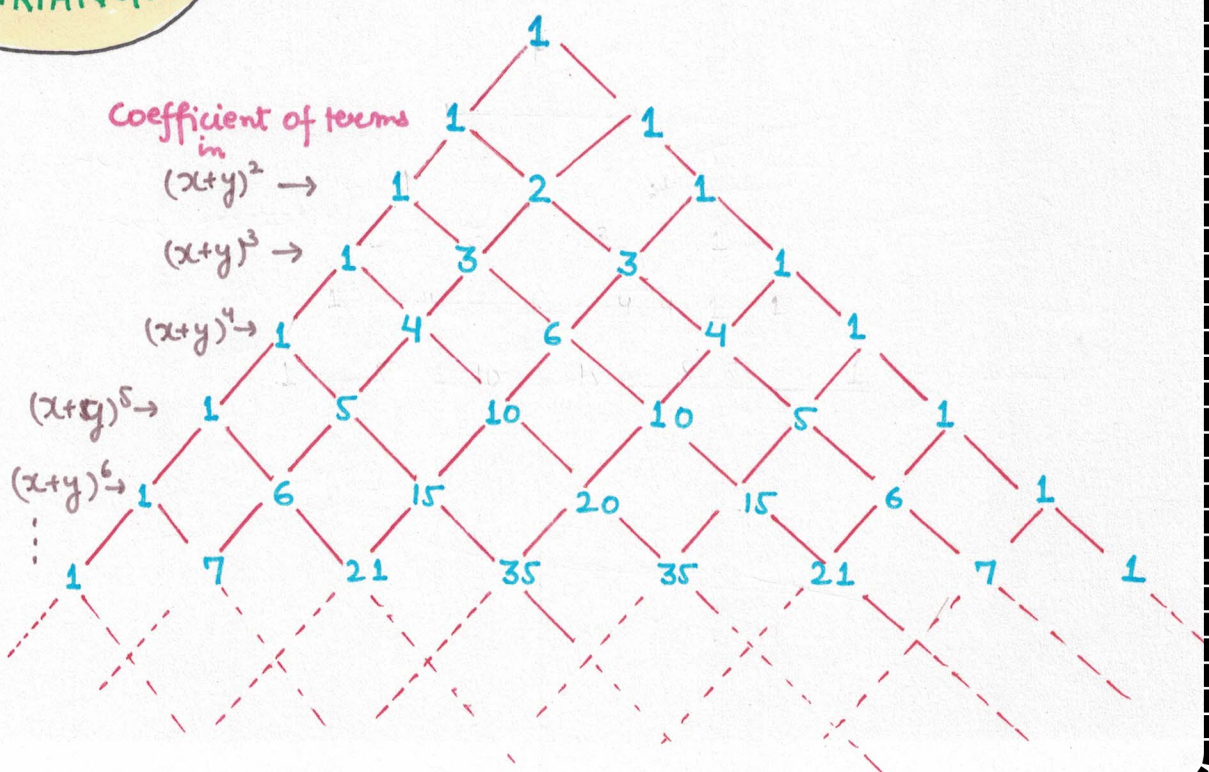
$(x+y)^2 = x^2+2xy+y^2$

$(x+y)^3 = x^3+3x^2y+3xy^2+y^3$

$(x+y)^4 = x^4+4x^3y+6x^2y^2+4xy^3+y^4$

$(x+y)^5 = x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$

### PASCAL'S TRIANGLE



# BINOMIAL THEOREM

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_{n-1} x y^{n-1} + {}^nC_n y^n$$

where, $${}^nC_0 = 1 = {}^nC_n$$

$${}^nC_r = \frac{n!}{(n-r)!\,r!}$$

$${}^nC_r = {}^nC_{n-r}$$

Ques: Find the expansion of $(x+y)^7$

Sol: $${}^nC_1 = {}^7C_1 = \frac{7!}{(7-1)! \times 1!} = \frac{6! \times 7}{6! \times 1} = 7$$

$${}^nC_2 = {}^7C_2 = \frac{7!}{(7-2)!\,2!} = \frac{5! \times 6 \times 7}{5! \times 2 \times 1} = 21$$

$${}^nC_3 = {}^7C_3 = \frac{7!}{(7-3)!\,3!} = \frac{4! \times 5 \times 6 \times 7}{4! \times 3 \times 2} = 35$$

$\Rightarrow {}^nC_4 = {}^nC_3$ ; ${}^nC_5 = {}^nC_2$ ; ${}^nC_6 = {}^nC_1$

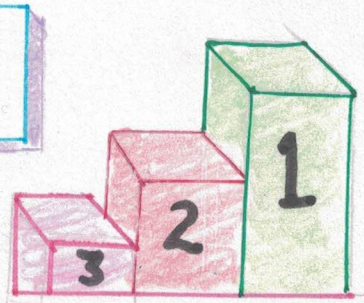
Hence,

$$(x+y)^7 = x^7 + 7x^6y^1 + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + y^7$$

Ques: Find the expansion of $\left(x - \frac{1}{x}\right)^4$

Sol: $$\left(x - \frac{1}{x}\right)^4 = x^4 - 4x^3\left(\frac{1}{x}\right) + 6x^2\left(\frac{1}{x}\right)^2 - 4x\left(\frac{1}{x}\right)^3 + \left(\frac{1}{x}\right)^4$$

$$\left(x - \frac{1}{x}\right)^4 = x^4 - 4x^2 + 6 - \frac{4}{x^2} + \frac{1}{x^4}$$



No. of terms in the expansion of $(x+y)^n$ is $(n+1)$

$T_r = {}^nC_{r-1}\, x^{n-r+1}\, y^{r-1}$

Ques: Find the 4th term in the expansion of $(x+y)^{20}$?

Sol: $n = 20 \qquad r = 4$

$T_r = {}^{20}C_3\, x^{17}\, y^3 \qquad [n-r+1 = 20-4+1 = 17]$

$${}^{20}C_3 = \frac{20!}{(20-3)!\,3!} = \frac{17! \times 18 \times 19 \times 20}{17! \times 3 \times 2} = 1140$$

$T_r = T_4 = 1140\, x^{17} y^3$

Ques: Find the coefficient of 6th term in the expansion of $\left(x - \frac{1}{x^2}\right)^9$ ?

Sol: $n = 9 \qquad r = 6$

$T_6 = {}^nC_{6-1}\, x^{n-6+1}\, y^{6-1}$

$T_6 = {}^9C_5\, x^4 y^5$

$${}^9C_5 = \frac{9!}{(9-5)! \times 5!} = \frac{4! \times 5 \times 6 \times 7 \times 8 \times 9}{4! \times 2 \times 3 \times 4 \times 5}$$

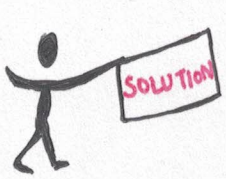
$= 126$

Coefficient of 6th term = 126

$T_6 = 126\, x^4 \left[-\frac{1}{x^2}\right]^5$

$= -126\, \frac{x^4}{x^{10}}$

$T_6 = -\frac{126}{x^6}$



$r^{th}$ term from end = $(n-r+2)^{th}$ term from beginning

Ques: Find the $5^{th}$ term from end of the expansion of $(x+y)^{20}$

Sol: $5^{th}$ term from end = $20-5+2$

$= 17^{th}$ term from beginning

$T_{17} = {}^{20}C_{17}\, x^3 y^{17}$

${}^{20}C_{17} = \dfrac{20!}{(20-17)!\,17!} = \dfrac{17! \times 18 \times 19 \times 20}{3 \times 2 \times 17!}$

$= 380 \times 3$

$\boxed{{}^{20}C_{17} = 1140}$

Ques: Find the coefficient of $x^{-8}$ in the expansion of $\left[x - \frac{1}{2x^2}\right]^{10}$

Sol: $x^{-8} = x^{n-r+1}\, y^{r-1}$

Here, $n = 10$

$x^{-8} = x^{11-r}\, y^{r-1}$

$x^{-8} = x^{11-r} \left[\dfrac{-1}{2x^2}\right]^{r-1}$

$x^{-8} = x^{11-r-2r+2} \left[\dfrac{1}{2^{r-1}}\right]$ → vanish it [of no use now]

$-8 = 13 - 3r$

$r = 7$

$\boxed{\text{Coefficient of } x^{-8} = \dfrac{{}^{10}C_6}{(2)^6} = \dfrac{105}{32}}$

Ques: Find the third term from the end in the expansion of $\left(x^3 - \frac{1}{x}\right)^9$

Sol: 3rd term from end = $(n-r+2)^{th}$ term from beginning

$= 9-3+2 = 8^{th}$

$T_8 = {}^9C_7\, x^2 y^7$

$= {}^9C_7 (x^3)^2 \left[-\frac{1}{x}\right]^7$

$= -{}^9C_7 \frac{x^6}{x^7} = -\frac{{}^9C_7}{x}$

$${}^9C_7 = \frac{9!}{(9-7)! \times 7!} = \frac{7! \times 8 \times 9}{2 \times 7!} = 36$$

$$\boxed{T_8 = -\frac{36}{x}}$$

Ques: Find the middle term of expansion $\left[ax - \frac{b}{x}\right]^9$

Sol: We know that expansion of $[x+y]^n$ has $(n+1)$ terms

So, $\left(ax - \frac{b}{x}\right)^9$ will have 10 terms

The expansion will have 2 middle terms:

$\left[\frac{10}{2}\right] = 5^{th}$ and $\left[\frac{10}{2}+1\right] = 6^{th}$ term

$$T_5 = {}^9C_4 (ax)^5 \left(\frac{-b}{x}\right)^4 = \frac{9!}{5! \times 4!} a^5 (b)^4 x$$

$$= \frac{6 \times 7 \times 8 \times 9}{2 \times 3 \times 4} a^5 (b)^4 x$$

$$\boxed{T_5 = 126 a^5 b^4 x}$$

$$T_6 = {}^9C_5 (ax)^4 \left(\frac{-b}{x}\right)^5 = \frac{9!}{4! 5!} a^4 x^4 \cdot \frac{(-b)^5}{x^5}$$

$$\boxed{T_6 = 126 \frac{a^4 b^5}{x}}$$



Ques: If $p$ and $q$ be positive, then prove that coefficients of $x^p$ and $x^q$ in the expansion of $(1+x)^{p+q}$ will be equal.

Sol: $x^p = (1)^{n-r+1}(x)^{r-1}$ $\qquad$ $x^q = (1)^{n-r+1}x^{r-1}$

$p = r-1$ $\qquad$ $q = r-1$

$r = p+1$ $\qquad$ $r = q+1$

$T_{p+1} = {}^nC_p \ldots.$ $\qquad$ $T_{q+1} = {}^nC_q \ldots.$

$${}^nC_p = \frac{(p+q)!}{(p+q-p)!\,p!} \qquad {}^nC_q = \frac{(p+q)!}{(p+q-q)!\,q!}$$

$$= \frac{(p+q)!}{p!\,q!} \qquad = \frac{(p+q)!}{p!\,q!}$$

Hence,

**Coefficient of $x^p$ = coefficient of $x^q$**

Ques: Find the term independent of $y$ in expansion of $(y^{-1/6} - y^{1/3})^9$

Sol: Term independent of $y$ will have $y^0$

$$y^0 = {}^nC_r[y^{-1/6}]^{n-r+1}[-y^{1/3}]^{r-1}$$

$$y^0 = {}^9C_r[y^{-1/6}]^{10-r}[y^{\frac{r-1}{3}}]$$

$$y^0 = y^{\frac{-10+r}{6} + \frac{r-1}{3}}$$

$$y^0 = y^{\frac{-10+r+2r-2}{6}}$$

$$0 = -12+3r$$

$$r = 4$$

$$T_4 = -{}^9C_3 = -\frac{9!}{3!\,6!} = -\frac{7\times 8\times 9}{3\times 2}$$

$$T_4 = -84$$

Ques: Find the coefficient of $x^5$ in the expansion of the product $(1+2x)^6(1-x)^7$.

Sol: Terms containing $x^5$ in product.

$$\text{coeff}_1\,(2x)^5 + \text{coeff}_2\,(2x)^4(-x) + \text{coeff}_3\,(2x)^3(-x)^2 + \text{coeff}_4\,(2x)^2(-x)^3 + \text{coeff}_5\,(2x)(-x)^4 + \text{coeff}_6\,(-x)^5$$

$$(1+2x)^6 = 1 + 6(2x)^5 + 15(2x)^4 + 20(2x)^3 + 15(2x)^2 + 6(2x) + (2x)^6$$

$$(1-x)^7 = 1 + 7(-x) + 21(-x)^2 + 35(-x)^3 + 35(-x)^4 + 21(-x)^5 + 7(-x)^6 + (-x)^7$$

$\text{Coeff}_1 = 6$ $\quad \text{Coeff}_2 = 15\times 7 = 105$ $\quad \text{Coeff}_3 = 20\times 21 = 420$

$\text{Coeff}_4 = 15\times 35 = 525$ $\quad \text{Coeff}_5 = 6\times 35 = 210$ $\quad \text{Coeff}_6 = 21$

Terms in $x^5 \rightarrow$ $6[2x]^5 + 105(2x)^4 x + 420(2x)^3 x^2 - 525(2x)^2 x^3 + 210(2x)(-x)^4 - 21x^5$

$$= 192x^5 - 1680x^5 + 8360x^5 - 2100x^5 + 4200x^5 - 21x^5$$

**Terms in $x^5$ = $171\,x^5$**

Ques: Find the middle term in the expansion of $\left[\frac{a}{x} + bx\right]^{12}$

Sol: No. of terms = 13

Middle term = $\frac{13+1}{2}$ = $7^{th}$ term

$T_7 = {}^{12}C_6\, x^6 y^6$

$$= \frac{12!}{6!\,6!}\left(\frac{a}{x}\right)^6 (bx)^6$$

$$= \frac{6!\times 7\times 8\times 9\times 10\times 11\times 12}{6!\times 2\times 3\times 4\times 5\times 6}\, a^6 b^6$$

**$T_7 = 924\,a^6 b^6$**

Ques: Find the middle term in the expansion of $\left[3x - \frac{x^3}{6}\right]^9$

Sol: No. of terms = 10

$5^{th}$ term and $6^{th}$ term → two middle terms

$$T_5 = {}^9C_4 x^5 y^4 = \frac{9!}{5! \times 4!} (3x)^5 \left[\frac{-x^3}{6}\right]^4$$

$$= \frac{6 \times 7 \times 8 \times 9}{2 \times 3 \times 4} \cdot 243 x^5 \cdot \frac{x^{12}}{2 \cdot 3}$$

$$= \frac{189}{8} x^{17}$$

$$T_6 = {}^9C_5 x^4 y^5 = \frac{9!}{5!\,4!} (3x)^4 \left(-\frac{x^3}{6}\right)^5$$

$$= \frac{6 \times 7 \times 8 \times 9}{2 \times 3 \times 4} \times \frac{3 x^{19}}{2 \times 3}$$

$$= -\frac{21 x^{19}}{16}$$

$$T_5 = \frac{189}{8} x^{17} \quad \text{and} \quad T_6 = -\frac{21}{16} x^{19}$$

Ques: Find the value of a if the $17^{th}$ and $18^{th}$ terms of $(2+a)^{50}$ are equal.

Sol: $T_{17} = {}^{50}C_{16} (2)^{34} (a)^{16}$

${}^{50}C_{16} = \frac{50!}{34!\,16!}$

$T_{18} = {}^{50}C_{17} (2)^{33} (a)^{17}$

$$\frac{50!}{34!\,16!} (a)^{16} = \frac{(2)^{33}\, 50!}{33!\,17!} (a)^{17}$$

$$\frac{2 \times 33! \times 17!}{33! \times 34 \times 16!} = a$$

$$\frac{16! \times 17}{16! \times 34} \times 2 = a \Rightarrow$$

$a = 1$

Ques: Find the $6^{th}$ term in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^{10}$

Sol: $T_6 = {}^{10}C_5 \ (2x^3)^5 \left(-\frac{1}{3x^2}\right)^5$

$$= -\frac{10!}{5!\,5!}(32\,x^{15})\left(\frac{1}{243\,x^{10}}\right)$$

$$= -\frac{5! \times 6 \times 7 \times 8 \times 9 \times 10}{5! \times 5 \times 4 \times 3 \times 2}\left[\frac{32}{243}\,x^5\right]$$

$$= -\frac{448 \times 2\,x^5}{27}$$

$$T_6 = -\frac{896}{27}x^5$$

## PROPERTIES OF BINOMIAL COEFFICIENT

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \ldots\ldots\ldots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$$

1. ${}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \ldots\ldots\ldots\ldots + {}^nC_{n-1} x^{n-1} + {}^nC_n = 2^n$

PROOF: We have $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots\ldots$

Put $x = 1$

$(1+1)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots$

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots$ Hence Proved

2. ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 + \ldots\ldots (-1)^{n-1} C_{n-1} + (-1)^n {}^nC_n = 0$

PROOF: We have $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots\ldots$

Put $x = -1$

$(1-1)^n = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 + \ldots\ldots$

$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 + \ldots\ldots$ Hence Proved

3 $\quad {}^nC_0 + {}^nC_2 + {}^nC_4 + {}^nC_6 + \ldots\ldots = 2^{n-1}$

PROOF: Adding property 1 and 2

$$\begin{aligned} {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + {}^nC_4 + \ldots\ldots &= 2^n \\ {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 + \ldots\ldots &= 0 \\ \hline 2\left[{}^nC_0 + {}^nC_2 + {}^nC_4 + {}^nC_6 + \ldots\ldots\right] &= 2^n \end{aligned}$$

or, $\quad {}^nC_0 + {}^nC_2 + {}^nC_4 + {}^nC_6 + \ldots\ldots = 2^{n-1}$ Hence Proved

4 $\quad {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots = 2^{n-1}$

PROOF: Subtract 2 from 1

$$\begin{aligned} {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots\ldots &= 2^n \\ (-)\ {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots\ldots\ldots &= 0 \\ \hline 2\left[{}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots\right] &= 2^n \end{aligned}$$

(+) (−) (+) (−)

Or $\quad {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots = 2^{n-1}$ Hence Proved.



Ques: Find ${}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + \ldots\ldots + {}^{10}C_{10}$

Sol: $2^n = 2^{10} =$ **1024**

Ques: Find ${}^8C_1 + {}^8C_3 + {}^8C_5 + {}^8C_7$

Sol: $2^{n-1} = 2^7 =$ **128**

Ques: Find ${}^{10}C_2 + {}^{10}C_4 + {}^{10}C_6 + {}^{10}C_8$

Sol: $2^{n-1} - {}^{10}C_0 + {}^{10}C_1 + {}^{10}C_{10}$

$= 2^9 - 1 - 1$

$= 512 - 2$

$=$ **510**

Ques: Find the value of $(\sqrt{2}+1)^6 - (\sqrt{2}-1)^6$

Sol: $(\sqrt{2}+1)^6 - (\sqrt{2}-1)^6$

$= (1+\sqrt{2})^6 - (1-\sqrt{2})^6$

$= {}^nC_0 + {}^6C_1(\sqrt{2}) + {}^6C_2(2) + {}^6C_3(2)^{3/2} + {}^6C_4(2)^2 + {}^6C_5(2)^{5/2} + {}^6C_6(2)^{6/2} - \left[{}^nC_0 - {}^6C_1(\sqrt{2}) + {}^6C_2(\sqrt{2})^2 - {}^6C_3(\sqrt{2})^3 + {}^6C_4(\sqrt{2})^4 - {}^6C_5(\sqrt{2})^5 + {}^6C_6(\sqrt{2})^6\right]$

$= 2\left[{}^6C_1(\sqrt{2}) + {}^6C_3(\sqrt{2})^3 + {}^6C_5(\sqrt{2})^5\right]$

$= 2[6!]$

$= 2\sqrt{2}\,[6 + 20\times2 + 6\times4]$

$= 2\sqrt{2}\,(70)$

$= \boxed{140\sqrt{2}}$

Ques: Find the value of ${}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots\ldots {}^{14}C_{11}$

Sol: ${}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots\ldots {}^{14}C_{13} = 2^{14-1} = 2^{13}$

$= {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots\ldots {}^{14}C_{11} = 2^{13} - {}^{14}C_{13}$

$= {}^{14}C_{13} = \dfrac{14!}{1!\,13!} = 14$

Sum $= 2^{13} - 14$

$= 8192 - 14 = \boxed{8178}$

Ques: Find the sum of the series $\sum_{r=0}^{10} {}^{20}C_r$

Sol: ${}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \ldots\ldots {}^{20}C_{10}$

$= \sum_{r=0}^{20} {}^{20}C_r - \sum_{r=10}^{20} {}^{20}C_r$

$= 2^{19} - [{}^{20}C_{11} + {}^{20}C_{12} + {}^{20}C_{13} + {}^{20}C_{14} + {}^{20}C_{15} + {}^{20}C_{16} + {}^{20}C_{17}]$

$= 2^{19} - \frac{1}{2}\sum_{10}^{20} {}^{20}C_r$

$= 2^{19} + \frac{1}{2}\,{}^{20}C_{10}$

$= {}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \ldots\ldots {}^{20}C_{20} = 2^{20}$

$= 2[{}^{20}C_0 + {}^{20}C_1 + \ldots\ldots {}^{20}C_{10}] = 2^{10} - {}^{20}C_{10}$

# APPLICATION OF BINOMIAL EXPANSION

## Finding Remainder of Binomial Theorem

Ques: Find the remainder when $3^{20}$ is divided by 8 ?

Sol:

$3^{20}$ (Nearest power of 3 to 8 = 9)

$\Downarrow$

$9^{10} = (1+8)^{10} \Rightarrow 1 + 8\,{}^{10}C_4 + 8^2\,{}^{10}C_4 + \ldots\ldots + 8^{10}\,{}^{10}C_{10}$

$1$ ↓ Remainder

$8\,{}^{10}C_4 + 8^2\,{}^{10}C_4 + \ldots\ldots + 8^{10}\,{}^{10}C_{10}$ — These all are divisible by 8

Ques: Find the remainder $2^{100}$ is divided by 15 ?

Sol:

$2^{100} = (2^4)^{25} = \underbrace{(15+1)^{25}}_{\text{Remainder} = 1}$

Ques: Find the remainder when $(13)^{900}$ is divided by 10.

Sol:

$(13)^{900} = (10+3)^{900}$

$= [3^2]^{450} = [3^4]^{9\times25}$

$= (10-1)^{450} = [100-19]^{25\times9}$

$= \underbrace{(1)^{450}}_{\text{Remainder } 1}$

OR

$= (169)^{450}$

$(170-1)^{450}$

Remainder = 1

Ques: Find the remainder when $5^{99}$ is divided by 13.

Sol: $(5^3)^{33} = (126)^{33} = (169-44)^{33}$

$= 5(25)^{49}$

$= 5[26-1]^{49}$

$= -5$

↳ Add 13 to this

$= \underbrace{8}_{\text{Remainder}}$

Alternatively,

$5[26K-1]$

$= 5 \times 26K - 5$

$= 13m - 5$

$= 13(m-1) + 13 - 5$ → Remainder cannot be -ve

$\Downarrow$

8

Ques: Find the remainder when $7^{103}$ is divided by 25

Sol: $7 \cdot 7^{102} = 7[49]^{51}$

$= 7[50-1]^{51} = 7[7 \times 2]^{51}$

$25 - 7 = 18$ } Remainder

Ques: Find the remainder when $5^{5^{5^{55\cdots}}}$ (24 times) is divided by 24

Sol: $5^{5 \times 5 \times 5 \cdots} = 5^{5^{24}} = 5^{(2K+1)}$

$= 5 \cdot 5^{2K} = 5 \cdot 25^{K}$

$= 5[24+1]^{K}$

$=$ Remainder $= 5$

Ques: Find the remainder when $2^{90} + 3^{90}$ is divided by 5.

Sol: $[4]^{45} + [9]^{45}$

$= [5-1]^{45} + [10-1]^{45}$

$= [-1]^{45} + [x]^{45} = -2$

$= -2 + 5 = 3$ → Remainder

Ques: If $9^7 - 7^9$ is divisible by $2^n$, then find the greatest value of $n$ $(n \in N)$

Sol: $[8+1]^7 - [8-1]^9$

$= 1 - [-1] = 2$

Remainder $= 0$

$\Rightarrow \frac{9^7}{2^n} - \frac{7^9}{2^n} = \frac{(8+1)^7}{2^n} - \frac{(8-1)^9}{2^n}$

$$= \frac{[2^3]^7}{2^n} - \frac{[2^3]^9}{2^n} + \frac{1}{2^n} + \frac{1}{2^n}$$

$$0 = 2^{21-n} - 2^{27-n} + 2^{1-n}$$

$$\frac{[8+1]^7}{2^n} - \frac{[8-1]^9}{2^n}$$

$n = 3$, Remainder = 2

$n = 2$, Remainder = 2

$$= \frac{8}{2^n} + \frac{1}{2^n}$$

Ques: Show that $9^{n+1} - 8n - 9$ is divisible by 64 where n is +ve integer

 HOME WORK

## Finding Unit Place digit:

Ques: Find unit place digit place digit of $[7]^{56}$

Sol: $[50-1]^{25} = [49]^{25}$ : 9 unit's place

Ques: Find 10's place digit in $7^{50}$

Sol: $50^2 - 50 + 1 = -49$ Term's place = 4

$$(50-1)^{25} = {}^{25}C_0 (50)^{50} - {}^{25}C_1 (50)^{49} + {}^{25}C_2 (50)^{48} + \ldots - {}^{25}C_{23}(50)^2 + {}^{25}C_{24} 50 - {}^{25}C_{25}$$

$$= 25 \times 50 - 1 = 1249$$

Term's place = 4

Ques: Find the ten's place, unit's place and 100's place digit in $3^{100}$

Sol: $9^{50} = [10-1]^{50}$ unit's place = 1

$$[10-1]^{50} = -{}^{50}C_{49}\, 10 - {}^{50}C_{20}$$

$$= \frac{50!}{49!} = 50 \times 10 - 1 \cdot 500 - 1 = 499$$

$$[10-1]^{50} = {}^{50}C_0\, 10 - {}^{50}C_1 (10)^2 + {}^{50}C_2 (10)^3 \ldots\ldots - {}^{50}C_{48}(10)^2 + {}^{50}C_{49}(10) - {}^{50}C_{50}$$

$= 50 \times 10 - 1 = 500 - 1 = 499$

$v \to 1 \quad t \to 0 \quad h \to 0$

## Greatest Integer Function

$[x]$ = greatest integer less than or equal to $x$

↓ box $x$

$[2.5] = 2 \quad [3.7] = 3 \quad [-1.2] = 2$

$[-1.7] + [1.7] = -2 + 1 = -1$

All no.'s have 2 parts (a) Integral (b) Fractional

$[x]$ gives you integral part of any number

$x = [x] + f$

$3.7 = 3 + 0.7$

↓ Integral part ↓ Fractional part

$[x]$ = Integral part = I

$\{x\}$ = Fractional part = f

Example → 4.2 $\quad I = [4.2] = 4 \quad f = \{4.2\} = 0.2$

→ −3.3 $\quad I = [-3.3] = -4 \quad f = \{-3.3\} = 0.7$

* $0 \le f \le 1$

Ques: Find integral part of $(2+\sqrt{3})^{20} = {}^{20}C_0$

Sol: $(2+\sqrt{3})^{10} + (2-\sqrt{3})^{10}$ } odd terms will cancel

$${}^{10}C_2 = \frac{10!}{8!\,2} = \frac{9 \times 10}{2} = 45$$

$${}^{10}C_3 = \frac{7!\,8 \times 9}{3 \times 2 \times 7!} = 12$$

$${}^{10}C_4 = \frac{10!}{6! \times 4 \times 3 \times 2} = 210$$

Hence,

$$= 4+4 + 10[2][\sqrt{3}]^2 - 10(2)[\sqrt{3}]^2 + 45[2][\sqrt{3}]^4 - 45[2][\sqrt{3}]$$
$$+ 12[2][\sqrt{3}] - 12[2][\sqrt{3}] + 210[2][\sqrt{3}]$$

$$= 2\left[2^{10} + {}^{10}C_2\, 2^8 3 + {}^{10}C_4\, 2^6 9 + {}^{10}C_6\, 2^4 27 + {}^{10}C_8\, 2^2 \cdot 81 + {}^{10}C_{10}\, 2^{243}\right]$$

$$= I$$

$$(2+\sqrt{3})^{10} + (2-\sqrt{3})^{10} = I$$

$$[x] + f + f_1 = I$$

$0 \le f < 1$ and $0 < f_1 < 1$ also, $0 < f + f_1 < 2$

Since $f + f_1$ is integer $\Rightarrow f_1 + f = 1$

So,

$$[x] + 1 = I$$
$$[x] = I - 1$$

## GREATEST COEFFICIENT OF BINOMIAL EXPANSION

$(x+y)^{11} \Rightarrow \dfrac{{}^{n}C_{n+1}}{2} = {}^{11}C_6$

$$= \frac{11!}{6!5!} = \frac{6! \times 7 \times 8 \times 9 \times 10 \times 11}{6! \times 5 \times 4 \times 3 \times 2}$$

$$= 66 \times 7 = 462$$

${}^{11}C_7$ and ${}^{11}C_6$

$$(x+y)^n = {}^{n}C_0 x^n + {}^{n}C_1 x^{n-1} y + \ldots\ldots {}^{n}C_n y^n$$

If $n$ = even — Greatest coeff. = ${}^{n}C_{n/2}$

If $n$ = odd — Greatest coeff. = ${}^{n}C_{\frac{n+1}{2}}$ & ${}^{n}C_{\frac{n+3}{2}}$

# GREATEST TERM IN THE EXPANSION OF $(1+ax)^n$

Find $\left|\frac{T_{r+1}}{T_r}\right| \geq 1$

$\Rightarrow \frac{T_{r+1}}{T_r} = \frac{{}^nC_r (ax)^r}{{}^nC_{r-1} (ax)^{r-1}} \geq 1$

$\Rightarrow \left|\frac{n!\, ax\, (r-1)!\, (n-r+1)!}{(n-r)!\, r!\, n!}\right| \geq 1$

$\Rightarrow \left|\frac{(n-r+1)\, ax}{r}\right| \geq 1$

$\Rightarrow \frac{n-r+1}{r} |ax| \geq 1$

$\Rightarrow \frac{n-r+1}{r} \geq \frac{1}{ax}$

$\Rightarrow nax + ax \geq r(1+ax)$

$\Rightarrow ax(1+n) \geq r(1+ax)$

$\Rightarrow \boxed{\frac{|ax|\,(1+x) \geq r}{(1+|ax|)}}$

? Find the greatest value of $r$ from (1)

$T_{r+1}$ is the greatest term

Ques: Find the greatest term of $\left(1+\frac{3}{2}x\right)^{10}$ where $x = \frac{3}{5}$

Sol: $r \leq \frac{(n+1)|ax|}{1+|ax|}$

$r \leq \frac{(11)\left[\frac{3}{2} \times \frac{3}{5}\right]}{1+\left[\frac{3}{2} \times \frac{3}{5}\right]} = \frac{11 \times 9}{10+9} = \frac{99}{19}$

$T_{r+1} = T_{\frac{99+19}{19}} = T_{\frac{108}{19}} = T_{5.68\ldots}$ } $T_6$ will give greatest term

$$T_6 = {}^{10}C_5 \left(\frac{3}{2}x\right)^5$$

$$= \frac{10!}{5! \times 5!} = \frac{5! \times 6^2 \times 7 \times 8^2 \times 9 \times 10}{5! \times 2 \times 3 \times 4 \times 5}$$

$$= 14 \times 18 = 252$$

$$T_6 = 252 \times \frac{3^5}{2^4} x^5$$

$$= \frac{63 \times 81 \times 3x}{8}$$

$$= \frac{15369}{8} \left[\frac{3}{5}\right]^5$$

## GREATEST TERM IN THE EXPANSION OF $(x+y)^n$

$$\frac{T_{r+1}}{T_r} = \frac{{}^nC_r x^{n-r} y^r}{{}^nC_{r-1} x^{n-r+1} y^{r-1}} = \frac{n-r+1}{r} \cdot \frac{y}{x}$$

$$\left|\frac{T_{r+1}}{T_r}\right| \geq 1 \quad \{\text{For greatest term}\}$$

$$\Rightarrow \left|\frac{n-r+1}{r} \times \frac{y}{x}\right| \geq 1$$

$$\Rightarrow \frac{n-r+1}{r} \geq \left|\frac{x}{y}\right|$$

$$\Rightarrow n-r+1 \geq r\left|\frac{x}{y}\right|$$

$$\Rightarrow (n+1) \geq r\left(1+\left|\frac{x}{y}\right|\right)$$

$$\Rightarrow r \leq \frac{n+1}{1+\left|\frac{x}{y}\right|}$$

Then find the greatest integer value of $r$

$$r = \left[\frac{n+1}{1+|x/y|}\right]$$

Now, $T_{r+1}$ is greatest term

When $\frac{n+1}{1+\left|\frac{x}{y}\right|}$ is integer, there are 2 greatest term, $T_r$ and $T_{r+1}$

Ques: Find the greatest term in the expansion of $\left(\frac{2}{3}x-\frac{5}{4}\right)^{10}$ where, $x=2$

Sol: $r=\left[\frac{11}{1+\left[\frac{2}{3}\times\frac{4}{-5}\right]}\right] = \left[\frac{11\times15}{15+16}\right] = 5$

$T_r$ and $T_{r+1}$

$T_5 = {}^{10}C_4\left(\frac{2}{3}\times 2\right)^6\left(\frac{-5}{4}\right)^4$

$= {}^{10}C_4\left[\frac{4}{3}\right]^6\left[\frac{5}{4}\right]^2$

$\left(2+\frac{3x}{8}\right)^{10}$ $T_r = 4^{th}$ greatest term $\rightarrow 4^{th}$

$3 \le r < 4$

$3 \le \left(\frac{11}{1+\frac{2}{3x}\times 8}\right) \le 4$

$3 \le \frac{11\times 3x}{3x+16} \le 4$

$3 \le \frac{33x}{3x+16} \le 4$

$\frac{33x}{3x+16} - 3 \ge 0$

$\frac{33x-9x-48}{3x+16} \ge 0$

$\frac{24x-48}{3x+16} \ge 0$

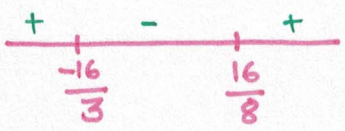
$\frac{8x-16}{3x+16} \ge 0$

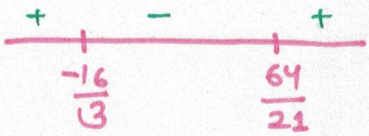
$\frac{33x}{3x+16} - 4 \le 0$

$\frac{33x-12x-64}{3x+16} \le 0$

$\frac{21x-64}{3x+16} \le 0$

$x = \frac{16}{8}$ $x = \frac{-16}{3}$ $x = \frac{64}{21}$ $x = \frac{-16}{3}$




Intersection → $2 < |x| < \frac{64}{21}$

$$\left(2, \frac{64}{21}\right) \cup \left(\frac{-64}{21}, -2\right)$$

## FINDING SUM OF BINOMIAL COEFFICIENT USING DIFFERENTIATION

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \ldots\ldots {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$

↳ Differentiate both sides with respect to $x$

$\Rightarrow n(1+x)^{n-1} \cdot 1 = 0 + {}^nC_1 + 2\,{}^nC_2 x + 3\,{}^nC_3 x^2 + \ldots\ldots n\,{}^nC_n x^{n-1}$

Now put $x = 1$

$\Rightarrow n(2)^{n-1} = {}^nC_1 + 2\cdot{}^nC_2 + 3\,{}^nC_3 + \ldots\ldots n\,{}^nC_n$

$$\boxed{{}^nC_1 + 2\cdot{}^nC_2 + 3\cdot{}^nC_3 + 4\,{}^nC_4 + \ldots\ldots n\,{}^nC_n = n(2)^{n-1}}$$

Now put $x = -1$

$n(1-n)^{n-1} = 0 + {}^nC_1 - 2\,{}^nC_2 + 3\,{}^nC_3 - 4\,{}^nC_4 + \ldots\ldots n\,{}^nC_n(-1)^{n-1}$

$$\boxed{{}^nC_1 - 2\,{}^nC_2 + 3\,{}^nC_3 - 4\,{}^nC_4 + \ldots\ldots n\,{}^nC_n(-1)^{n-1} = 0}$$

**Illustration:**

$(1+x)^{2n+1} = {}^{2n+1}C_0 + {}^{2n+1}C_1 (x)^1 + {}^{2n+1}C_2 (x)^2 + \ldots\ldots {}^{2n+1}C_{2n+1} (x)^{2n+1}$

Differentiate both sides with respect to $x$.

Put $x = -1$ $(2n+1)(1-1)^{2n} = 0 + {}^{2n+1}C_1 + 2\,{}^{2n+1}C_2(-1) + \ldots\ldots$

$2n+1\ C_{2n+1}\ (-1)^{2n}$

# USE OF INTEGRATION IN FINDING SUM OF BINOMIAL CO-EFFICIENTS

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots\dots {}^nC_n x^n$$

Integrating both sides with respect to $x$

$$\int_0^1 (1+x)^n dx = \int_0^1 ({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots {}^nC_n x^n) dx$$

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \left[{}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots \frac{{}^nC_n x^{n+1}}{n+1}\right]_0$$

$$\frac{(1+1)^{n+1}}{n+1} = {}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + \dots\dots \frac{{}^nC_n}{n+1}$$

$$\frac{2^{n+1}}{n+1} - \frac{1}{n+1} = {}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + \dots\dots \frac{{}^nC_1}{n+1}$$

$$\boxed{\frac{2^{n+1}-1}{n+1} = {}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + \dots\dots \frac{{}^nC_1}{n+1}}$$

Ques: Prove that:

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots\dots\dots\dots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$$

Sol: We have.

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^a = a C_0 + a^2 \frac{C_1}{2} + a^3 \frac{C_2}{3} + \dots\dots\dots$$

Put $x = 1 = a$

$$\boxed{\frac{2^{n+1}}{n+1} = C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots\dots\dots\dots + \frac{C_n}{n+1}}$$

Ques: $C_r = {}^nC_r$ Prove that

$$\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \dots\dots\dots = \frac{2^{n-1}}{n+1}$$

Sol: $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \frac{C_3}{4} + \dots\dots \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$ — (1)

$$(1-x)^n = C_0 - C_1 x + C_2 x^2 - C_3 x^3 + \dots\dots (-1)^n C_n x^n$$

$$\int_0^1 (1-x)^n dx = \int_0^1 [C_0 - C_1 x + C_2 x^2 - C_3 x^3 + \dots (-1)^n C_n x^n] dx$$

$$\Rightarrow \quad 1-x = t \quad \Rightarrow \quad -1 = \frac{dt}{dx} \quad \Rightarrow dx = -dt$$

$$\int_1^0 t^n dt = \int_0^1 t^n dt = \left[\frac{t^{n+1}}{n+1}\right]_0^1$$

$$= \frac{1^{n+1}}{n+1} = \frac{1}{n+1}$$

NOW,

$$\frac{1}{n+1} = \int_0^1 \left[C_0 x - \frac{C_1 x^2}{2} + \frac{C_2 x^3}{3} - \frac{C_3 x^4}{4} + \dots \frac{(-1)^n C_n x^{n+1}}{n+1}\right]$$

$$\Rightarrow \frac{1}{n+1} = C_0 - \frac{C_1}{2} + \frac{C_2}{3} + \frac{C_3}{4} + \dots (-1)^n \frac{C_n}{n+1}$$

$$\Rightarrow \frac{1}{n+1} = C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} \dots \quad (2)$$

EQUATION 1 - EQUATION 2

$$2\left[\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \dots\right] = \frac{2^{n+1}-1}{n+1} - \frac{1}{n+1}$$

$$2\left[\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \dots\right] = \frac{2^{n+1}-1-1}{n+1}$$

$$\boxed{\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \dots = \frac{2^n - 1}{n+1}}$$

Ques: $C_0 + \frac{C_1}{2}x + \frac{C_2}{3}x^2 + \dots + \frac{C_n}{n+1}x^n = \frac{(1+x)^{n+1}-1}{(n+1)x}$

Sol: $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots C_n x^n$

Integrate both sides with respect to $x$

$$\Rightarrow \int \frac{(1+x)^n}{x} dx = \int \frac{C_0 + C_1 x + C_2 x^2 + \dots C_n x^n}{x} \cdot dx$$

$$\Rightarrow \int \left(\frac{1}{x}\right)^n dx + \int x^{n-1} dx = \int \left(\frac{C_0}{x} + C_1 + C_2 x + C_3 x^2 + \dots C_n x^{n-1}\right) dx$$

# SERIES CONTAINING PRODUCT OF BINOMIAL CO-EFFICIENT

$C_0C_0 + C_1C_1 + C_2C_2 + \dots + C_nC_n = ?$

$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ —— ①

$(x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n$ —— ②

Multiply 1 & 2 -

$(1+x)^n(x+1)^n = (C_0 + C_1x + \dots + C_nx^n)(C_0x^n + C_1x^{n-1} + \dots + C_n)$

Coefficient of $x^n$ from RHS -

$C_0C_0 + C_1C_1 + C_2C_2 + \dots + C_nC_n$

Coefficient of $x^n$ from LHS -

$(1+x)^n(1+x)^n = (1+x)^{2n}$

Coefficient of $x^n$ in $(1+x)^{2n}$

$^{2n}C_n$

$$C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2 = {}^{2n}C_n$$

Ques: $C_0C_1 + C_1C_2 + C_2C_3 + \dots + C_{n-1}C_n = \frac{(2n)!}{(n-1)!(n+1)}$

Sol:

$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$

$(x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + C_3x^{n-3} + \dots + C_n$

RHS Coefficient of $x^{n-1}$

$= C_0C_1 + C_1C_2 + C_2C_3 + C_3C_4 + \dots + C_{n-1}C_n$

LHS Coefficient of $x^{n-1}$

$(1+x)^{2n} \longrightarrow {}^{2n}C_{n-1} = \frac{(2n)!}{(2n-n+1)!(n-1)!}$

$= \frac{2n!}{(n+1)!(n-1)!}$

LHS = RHS

Ques: $C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots\dots\dots + C_{n-r} C_n = \frac{(2n)!}{(n-r)!\,(n+r)!}$

Sol: $(1+x)^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots\dots + C_n x^n$

$(x+1)^n = C_0 x^n + C_1 x^{n-1} + C_2 x^{n-2} + \dots\dots + C_n$

Multiply them.

RHS Coeff. of $x^{n-r} = C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots\dots C_{n-r} C_n$

LHS Coeff. of $x^{n-r}$ in $(1+x)^{2n} = {}^{2n}C_{n-r}$

$$= \frac{2n!}{(2n-n+r)!\,(n-r)!}$$

$$= \frac{2n!}{(n+r)!\,(n-r)!}$$

Hence Proved

Ques: ${}^{n-1}C_{n-1}\,{}^nC_1 + {}^{n-1}C_{n-2}\,{}^nC_2 + {}^{n-1}C_{n-3}\,{}^nC_3 + \dots\dots + {}^{n-1}C_0\,{}^nC_n$

$= {}^{2n-1}C_{n-1}$

Sol: $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots\dots\dots + {}^nC_n x^n$

$(x+1)^{n-1} = {}^{n-1}C_0 x^{n-1} + {}^{n-1}C_1 x^{n-2} + {}^{n-1}C_2 x^{n-3} + \dots\dots\dots$

Multiply them,

RHS Coeff. of $x^{n-1} = {}^{n-1}C_{n-1}\,{}^nC_1 + {}^{n-1}C_{n-2}\,{}^nC_2 + {}^{n-1}C_{n-3}\,C_3 + \dots {}^{n-1}C_0\,{}^nC_n$

LHS Coeff. of $x^{n-1} = (1+x)^n (1+x)^{n-1} = (1+x)^{2n-1}$

Coefficient $= {}^{2n-1}C_{n-1}$

Ques: ${}^{n-1}C_0\,{}^nC_2 + {}^{n-1}C_1\,{}^nC_3 + {}^{n-1}C_2\,{}^nC_4 + \dots\dots\dots + {}^{n-1}C_{n-2}\,{}^nC_n = {}^{2n-1}C_{n-2}$

Sol: $(1+x)^{n-1} = {}^{n-1}C_0 + {}^{n-1}C_1 x + {}^{n-1}C_2 x^2 + \dots\dots\dots {}^{n-1}C_{n-1} x^n$

$(x+1)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} + \dots\dots {}^nC_n x^{n-2}$

Multiplying both,

Coeff. of $x^{n-2}$ in RHS: ${}^{n-1}C_0\,{}^nC_2 + {}^{n-1}C_1\,{}^nC_3 + {}^{n-1}C_2\,{}^nC_n + \dots {}^{n-1}C_{n-2}\,{}^nC_n$

— (1)

Coeff. of $x^{n-2}$ in LHS: $(1+x)^{n-1}(1+x)^n = (1+x)^{n-1}$

Coefficient = $^{2n-1}C_{n-2}$ — ②

1 = 2 Hence Proved

Ques: $^{n-1}C_{n-1}\,{}^nC_1 + {}^{n-1}C_{n-2}\,{}^nC_2 + {}^{n-1}C_{n-3}\,{}^nC_3 + \dots + {}^{n-1}C_0\,{}^nC_n = {}^{2n-1}C_{n-1}$

Sol:

$(1+x)^n = {}^nC_0x^n + {}^nC_1x^{n-1} + {}^nC_2x^{n-2} + {}^nC_3x^{n-3} \dots {}^nC_n$

$(x+1)^{n-1} = {}^{n-1}C_0x^{n-1} + {}^{n-1}C_1x^{n-2} + {}^{n-1}C_2x^{n-3} + \dots {}^{n-1}C_{n-1}x^0$

Multiply them,

Coeff. of $x^{n-1}$ RHS: $^{n-1}C_{n-1}\,{}^nC_1 + {}^{n-1}C_{n-2}\,{}^nC_2 + \dots {}^{n-1}C_0\,{}^nC_n$ — ①

Coeff. of $x^{n-1}$ LHS: $(x+1)^n(x+1)^{n-1} = (x+1)^{2n-1}$

Coefficient = $^{2n-1}C_{n-1}$ — ②

1 = 2 Hence Proved

## BINOMIAL THEOREM FOR RATIONAL POWER

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{1\times2\times3}x^3 + \frac{n(n-1)(n-2)(n-3)}{1\times2\times3\times4}x^4 + \dots$$

When $|x|<0 \Rightarrow -1<x<1$

and n is a rational index

This expansion will contain infinite terms, it will never vanish.

- $(1-x)^{-1} = 1 + x + \frac{(-1)(-2)}{2}x^2 + \frac{(-1)(-2)(-3)}{6}(-x)^3 + \frac{(-1)(-2)(-3)(-5)}{24}(-x)^4 + \dots$

$= 1 + x + x^2 + x^3 + \dots\dots$

$= \frac{1(1-x^n)}{1-x} = \frac{1}{1-x}$

- $(1-x)^{-2} = 1 + 2x + \frac{(-2)(-3)(-x)^2}{2} + \frac{(-2)(-3)(-4)(-x)^3}{1\times2\times3}$

$+ \frac{(-2)(-3)(-4)(-5)x^4}{1\times2\times3\times4} + \dots\dots$

$= 1 + 2x + 3x^2 + 4x^2 + \dots\dots$

$|x| < 1$

- $(1-x)^{-3} = 1 + 3x + \frac{(-3)(-4)(-x)^2}{1\times2\times3} + \dots$

$= 1 + 3x + 6x^2 + 10x^3 + 15x^4 + \dots$

Ques: Find the coeff. of $x^{10}$ in the expansion of $(1-x)^{30}$.

Sol: $(1-x)^{-30} = 1 + 30x + \frac{(-30)(-31)(-x)^2}{2} + \frac{(-30)(-31)(-32)(-x)^3}{6}$

$x^{10}$ coeff $\rightarrow$ $\frac{-30x-31x-32x-33x-34x-35x-36x-37}{1\times2\times3\times4\times5\times6\times7\times8\times9\times10}$

$x^5$ coeff $\rightarrow$ $\frac{-30x-31x-32x-33x-34x^5}{1\times2\times3\times4\times5}$

$= \frac{34!}{5!\,29!} = {}^{34}C_5$

$$(1-x)^{-n} = 1 + {}^nC_1 x + {}^{n+1}C_2 x^2 + {}^{n+2}C_3 x^3 + \dots\dots$$

## APPROXIMATION

When $x$ is very small, the highest power of $x$ in the expansion of $(1+x)^{-n}$ becomes negligible.

$$(1-x)^{-n} = 1 + {}^{n}C_1 x + {}^{n+1}C_2 x^2 + \underbrace{{}^{n+2}C_3 x^3 + {}^{n+3}C_4 x^4 + \dots\dots\dots}_{\text{Negligible}}$$

$$(1-x)^{-n} \simeq 1 + {}^{n}C_1 x + {}^{n+1}C_2 x^2$$

Or also,

$$(1-x)^{-n} = 1 - nx + \frac{n(n-1)}{2}x^2 - \underbrace{\frac{n(n-1)(n-2)}{1\times 2\times 3}x^3 + \dots\dots\dots}_{\text{Negligible}}$$

$$(1-x)^{n} \simeq 1 - nx + \frac{n(n-1)}{1.2}x^2$$

**Ques:** Find the approximate value of :-

a) $\sqrt[3]{0.99} = (1 - 0.01)^{1/3}$

$$= 1 - \frac{1}{3}(0.01) + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{1.2}(0.01)^2$$

$$= 1 - 0.003 - \frac{(-2)(0.0001)}{3.6 \times 3}$$

$$= 0.9970 - 0.0001$$

$$= 0.9969$$

b) $\left(\frac{1}{0.9}\right)^5$ $= 1 - x = \frac{10}{9} \Rightarrow 1 - \frac{10}{9} = x \Rightarrow x = -\frac{1}{9}$

$$= \left(1 + \frac{1}{9}\right)^5$$

$$= 5\left(\frac{1}{9}\right)$$

$$= (1-0.1)^{-5} = 1 + {}^{5}C_1 (0.1) + {}^{6}C_2 (0.1)^2$$

$$= 1 + 0.5 + 3\times 4\times 5(0.01)$$

$$= 1 + 0.5 + 0.15$$

$$= 1.65$$

## SOME MORE EXPANSIONS.

[I] $e^x = 1 + \frac{x}{1} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\dots\dots\dots$ $\infty$ term

$$e^{-x} = 1 + \frac{-x}{1} + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \frac{(-x)^4}{4!} + \dots\dots \infty \text{ term}$$

[II] $$e^{-x} = 1 - \frac{x}{1} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots\dots \infty \text{ term}$$

[III] $$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} \dots\dots \infty \text{ term}$$

[IV] $$\log_e(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \dots\dots \infty \text{ term}$$

Ques: Find the value of $1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\dots$ $\infty$ in terms of e.

Sol: $$e^1 = 1 + \frac{1}{1} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \dots\dots$$

$$e^{-1} = 1 - \frac{1}{1} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \dots\dots$$

$$e + e^{-1} = 2\left[1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\right]$$

$$\frac{e+e^{-1}}{2} = 1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} \dots\dots$$

Ques: Find the approximate value of e (3 decimal)

Sol: $$e^1 = 1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{2.3} + \frac{1}{2.3.4} + \frac{1}{2.3.4.5}$$

$$e = 2 + 0.5 + 0.16 + 0.05 + 0.007$$

$$e = 2 + 0.717 = 2.717$$

Exact value of e = 2.718281828

Ques: Find the value of:

a) $$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$$

$$\log_e(1+1) = \log_e 2$$

b) $$\log_e 2 = e^2 = (2.71)^2$$

$$= \boxed{0.6987}$$

c) $\ln 10 = \log_e 10$

$= (2.718)^{10}$

$= (2 + 0.71)^{10}$

$= 2^{10}(1.358)^{10}$

$= 1024\,(10 + 1 + {}^{10}C_2(0.358) + {}^{10}C_3(0.358)$

$= 1024\,(3.58 + 36(0.358))$

$= \boxed{2.303}$